

CHAPTER-1

INTRODUCTION

Flood alerting and rescue systems are vital components in safeguarding communities from the destructive effects of flooding. These systems leverage advanced technologies, including sensors, satellite imagery, and real-time data analytics, to monitor water levels and weather conditions, enabling accurate predictions of potential flood events. By issuing timely alerts through various communication channels—such as mobile applications, SMS, and social media—these systems ensure that residents and authorities are informed well in advance, allowing for proactive measures to be taken.

In addition to alerting the public, flood alerting systems play a crucial role in coordinating emergency response efforts. They facilitate the organization of rescue operations, ensuring that emergency services can respond swiftly and effectively to those in need. This includes identifying safe evacuation routes and deploying resources where they are most needed.

Furthermore, community engagement and education are essential aspects of these systems, fostering a culture of preparedness and resilience among residents. By raising awareness of flood risks and response protocols, these systems not only save lives but also minimize economic losses. Overall, flood alerting and rescue systems are indispensable tools in building safer, more resilient communities in the face of increasingly frequent and severe flooding events.

1.1 OBJECTIVE OF PROJECT

The objective of flood alerting and rescue systems is to enhance community safety and resilience by providing timely warnings and effective response strategies in the face of flooding events. These systems aim to detect potential flooding risks early, ensuring that affected populations receive accurate information about the severity of the threat and recommended actions.

Additionally, these systems focus on optimizing resource allocation, educating communities on flood preparedness, and supporting post-flood recovery efforts. both people and property while fostering greater awareness and preparedness for future events. These systems aim to detect potential flooding risks early. both people and property while fostering greater awareness and preparedness for future events.

CHAPTER-2

LITERATURE OF SURVEY

1 Abdullahi, S. I., Habaebi, M. H., Gunawan, T. S., & Islam, M. R. Miniaturized Water Flow and Level Monitoring System for Flood Disaster Early Warning.

This paper presents the performance of a prototype miniaturised water flow and water level monitoring sensor designed towards supporting flood disaster early warning systems. The design involved selection of sensors, coding to control the system mechanism, and automatic data logging and storage. During the design phase, the apparatus was constructed where all the components were assembled using locally sourced items. Subsequently, under controlled laboratory environment, the system was tested by running water through the inlet during which the flow rate and rising water levels are automatically recorded and stored in a database via Microsoft Excel using Coolterm software. The system is simulated such that the water level readings measured in centimeters is output in meters using a multiplicative of 10. A total number of 80 readings were analyzed to evaluate the performance of the system. The result shows that the system is sensitive to water level rise and yielded accurate measurement of water level. But, the flow rate fluctuates due to the manual water supply that produced inconsistent flow. It was also observed that the flow sensor has a duty cycle of 50% of operating time under normal condition which implies that the performance of the flow sensor is optimal.

2 Billa, L., Mansor, S., & Rodzi Mahmud, A. Spatial information technology in flood early warning systems.

Malaysia experiences a major flood event every three years due to the adverse effects of two monsoon seasons a year. Floods have thus become the most significant natural disaster in the country in terms of the population affected, frequency, aerial extent, financial cost and the disruption to socio-economic activities. Many previous flood control measures have had different levels of success but have generally had little effect in reducing the problem. However, it is now understood that it is neither possible nor desirable to control floods completely. Spatial information technology is thus being increasingly recognized as the most effective approach to flood disaster management. This paper reviews the spatial information technology in flood disaster management and its application in Malaysia. Some flood forecasting systems are discussed, along with their shortcomings. The paper discusses the

framework of a proposed flood early warning system for the Langat river basin that operationally couples real-time NOAA-AVHRR data for quantitative precipitation forecasting with hydrologically oriented GIS and a MIKE11 hydrodynamic model.

3 Satria, D., Yana, S., Munadi, R., & Syahreza, S. Prototype of Google Maps-Based Flood Monitoring System Using Arduino and GSM Module.

The purpose of this paper is to develop prototype of the information system of the flood monitoring based internet of things (IoT). This prototype serves to assist users in accessing flood levels through water levels and rainy weather conditions. This paper presents the design of information system of flood monitoring based internet of things (IoT). This prototype study acquires water level and rainfall data using ultrasonic sensors HC-SR04 and rain sensor. Data of flood height and rain levels detected by sensors are processed using Arduino Uno Microcontroller to produce output data in HTML format. Flood altitude information system and rainy weather from the microcontroller are distributed using ethernet module as web server integrated with Wireless N Router TL-MR3020 as a gateway path to the user.

2.1 Existing System:

Existing flood alerting and rescue systems typically use a combination of water level sensors, meteorological data, and communication networks to monitor and predict flood risks. Advanced systems often employ radar and ultrasonic sensors in rivers and reservoirs to measure rising water levels, while satellite and weather data provide information on rainfall and upstream river flow. In urban settings, water level sensors are installed in drainage systems and flood-prone areas to detect sudden surges. These systems usually integrate with remote monitoring centers, where real-time data from sensors are analyzed to predict potential flooding. When critical levels are reached, automated alerts are issued via text messages, sirens, or mobile app notifications to residents and local authorities. Some systems use GSM, radio, or internet-based communication channels to ensure reliable data transmission, even in remote areas. Higher-end systems may incorporate machine learning algorithms to predict flood patterns, enabling authorities to optimize rescue and evacuation plans. While effective, these systems can be costly and complex, requiring extensive infrastructure and technical support. These systems usually integrate with remote monitoring centers, where real-time data from sensors are analyzed to predict potential flooding.

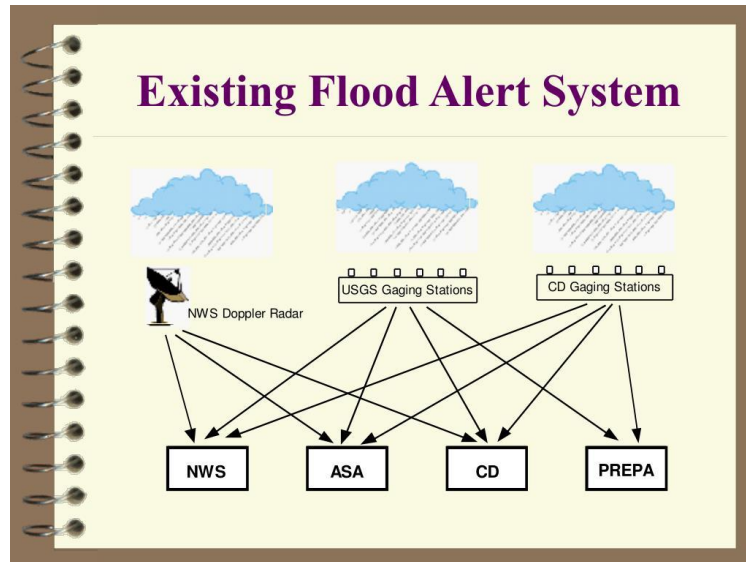


Fig 2.1 Existing flood alerting system

2.2 Proposed System:

The aim of this project is to develop a certain system which is efficient enough to predict the weather conditions and level of the water in water reservoirs so that preventive measures can be put in place priorly. This project is based on the open source electronic platform i.e. Arduino. The Arduino Uno R3 is to be set up multiple different devices such as water level sensor for the water level detection by capturing time between transmitting and receiving values, temperature and humidity sensor DHT11 for analyzing the moisture conten.

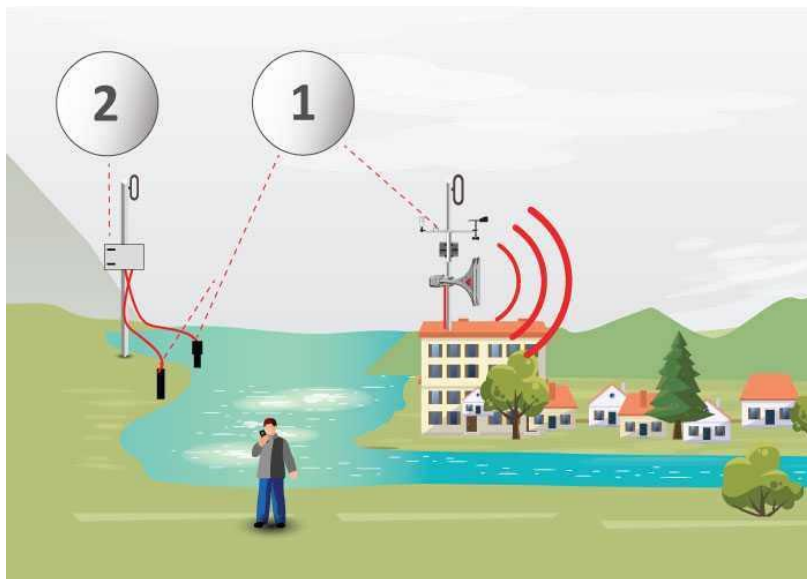
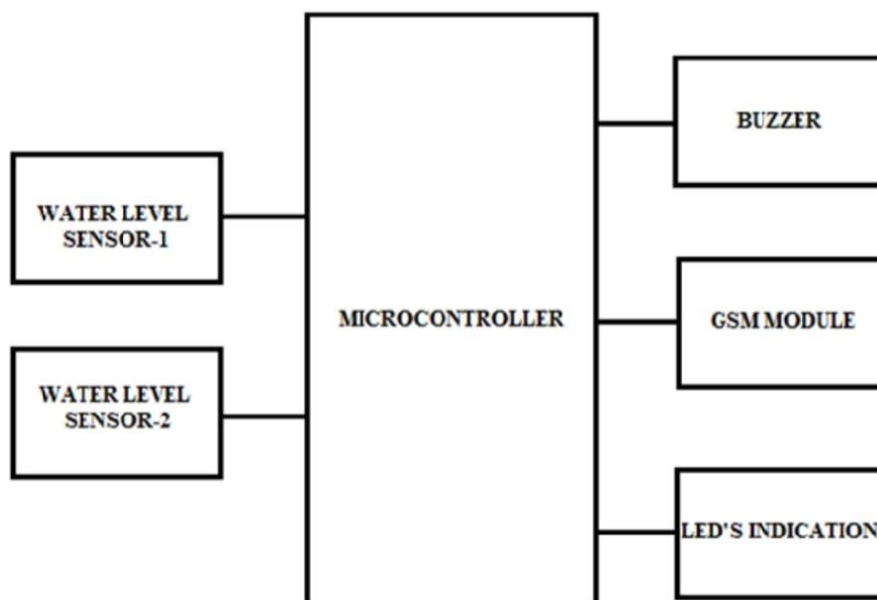


Fig 2.2 Proposed system

CHAPTER-3

HARDWARE DESCRIPTION

Block diagram:



List of modules:

- 1) Power Supply
- 2) Arudino uno
- 3) Water level sensor
- 4) Buzzer
- 5) LED's
- 6) GSM

3.1 Arduino UNO:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of To do so

you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

3.1.1 ATMEGA 328P (ARDUINO UNO)

The Arduino Uno is a microcontroller board based on the ATmega328. It has a 16 MHz ceramic resonator, 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a USB connection, a power jack, an ICSP header, and a reset button. This board is very simple and can be easily used, everything you need to support the microcontroller is in this board, just plug it in a computer via USB cable and power using an AC-to-DC adapter or battery to get started.

The difference seen in the Arduino Uno is that it does not use the FTDI USB-to-serial driver chip but, it has the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB to-serial converter. The ATmega328 is one kind of single-chip microcontroller formed with Atmel within the megaAVR family. The architecture of this Arduino Uno is a customized Harvard architecture with 8 bit RISC processor core. Other boards of Arduino Uno include Arduino Pro Mini, Arduino Nano, Arduino Due, Arduino Mega, and Arduino Leonardo.

Arduino Uno can detect the surroundings from the input. Here the input is a variety of sensors and these can affect its surroundings through controlling motors, lights, other actuators, etc. The ATmega328 is one kind of single-chip microcontroller formed with Atmel within the megaAVR family. The architecture of this Arduino Uno is a customized Harvard

architecture with 8 bit RISC processor core. Arduino Mega, and Arduino Leonardo. Arduino Uno can detect the surroundings from the input.

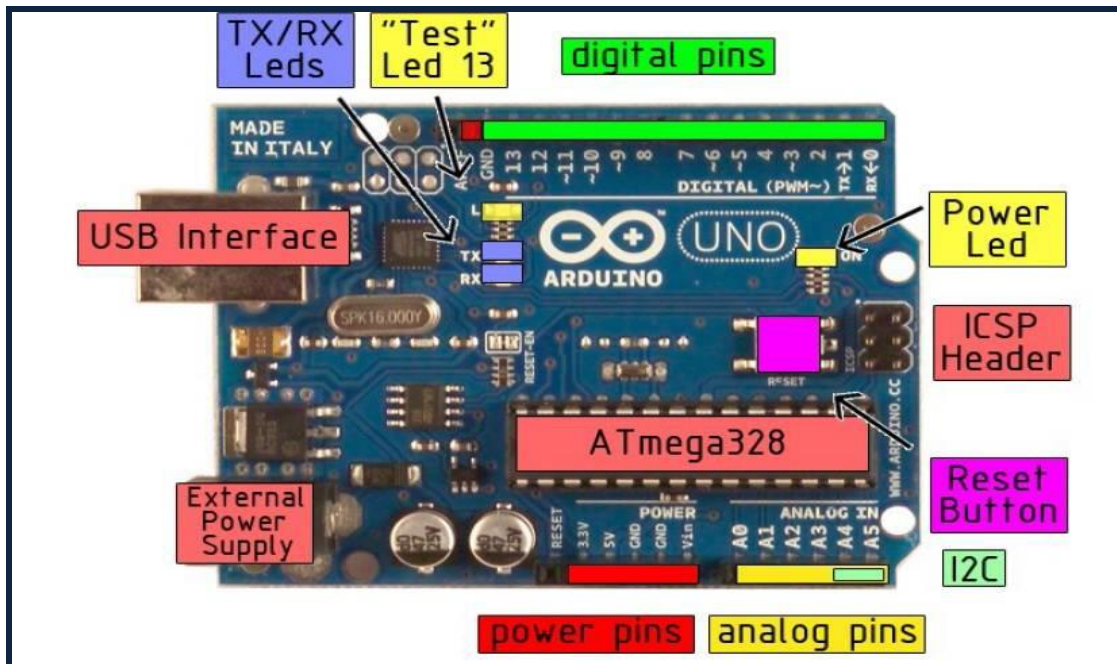


Fig 3.1: Arduino-Uno Structure

3.1.2 PIN DESCRIPTION:

(PCINT14/RESET) PC6	1	28	PC5 (ADC5/SCL/PCINT13)
(PCINT16/RXD) PD0	2	27	PC4 (ADC4/SDA/PCINT12)
(PCINT17/TXD) PD1	3	26	PC3 (ADC3/PCINT11)
(PCINT18/INT0) PD2	4	25	PC2 (ADC2/PCINT10)
(PCINT19/OC2B/INT1) PD3	5	24	PC1 (ADC1/PCINT9)
(PCINT20/XCK/T0) PD4	6	23	PC0 (ADC0/PCINT8)
VCC	7	22	GND
GND	8	21	AREF
(PCINT6/XTAL1/TOSC1) PB6	9	20	AVCC
(PCINT7/XTAL2/TOSC2) PB7	10	19	PB5 (SCK/PCINT5)
(PCINT21/OC0B/T1) PD5	11	18	PB4 (MISO/PCINT4)
(PCINT22/OC0A/AIN0) PD6	12	17	PB3 (MOSI/OC2A/PCINT3)
(PCINT23/AIN1) PD7	13	16	PB2 (SS/OC1B/PCINT2)
(PCINT0/CLKO/ICP1) PB0	14	15	PB1 (OC1A/PCINT1)

Fig 3.2: Pin Description

VCC

Digital supply voltage.

GND

Ground.

Port B (PB7:0) XTAL1/XTAL2/TOSC1/TOSC2

Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running. Depending on the clock selection fuse settings, PB6 can be used as input to the inverting Oscillator amplifier and input to the internal clock operating circuit. Depending on the clock selection fuse settings, PB7 can be used as output from the inverting Oscillator amplifier. If the Internal Calibrated RC Oscillator is used as chip clock source, PB7..6 is used as TOSC2..1 input for the Asynchronous Timer/Counter2 if the AS2 bit in ASSR is set.

The various special features of Port B are elaborated in table

The alternate pin configuration is as follows:

• XTAL2/TOSC2/PCINT7 – Port B, Bit 7

XTAL2: Chip clock Oscillator pin 2. Used as clock pin for crystal Oscillator or Low-frequency crystal Oscillator. When used as a clock pin, the pin can not be used as an I/O pin.

TOSC2: Timer Oscillator pin 2. Used only if internal calibrated RC Oscillator is selected as chip clock source, and the asynchronous timer is enabled by the correct setting in ASSR.

When the AS2 bit in ASSR is set (one) and the EXCLK bit is cleared (zero) to enable asynchronous clocking of Timer/Counter2 using the Crystal Oscillator, pin PB7 is disconnected from the port, and becomes the inverting output of the Oscillator amplifier. In this mode, a crystal Oscillator is connected to this pin, and the pin cannot be used as an I/O pin.

PCINT7: Pin Change Interrupt source 7. The PB7 pin can serve as an external interrupt source. If PB7 is used as a clock pin, DDB7, PORTB7 and PINB7 will all read 0.

PCINT6: Pin Change Interrupt source 6. The PB6 pin can serve as an external interrupt source. The PB7 pin can serve as an external interrupt source.

If PB7 is used as a clock pin, DDB7, PORTB7 and PINB7 will all read 0. Timer Oscillator pin 2.

• XTAL1/TOSC1/PCINT6 – Port B, Bit 6

XTAL1: Chip clock Oscillator pin 1. Used for all chip clock sources except internal calibrated RC Oscillator. When used as a clock pin, the pin can not be used as an I/O pin.

TOSC1: Timer Oscillator pin 1. Used only if internal calibrated RC Oscillator is selected as chip clock source, and the asynchronous timer is enabled by the correct setting in ASSR. When the AS2 bit in ASSR is set (one) to enable asynchronous clocking of Timer/Counter2, pin PB6 is disconnected from the port, and becomes the input of the inverting Oscillator amplifier. In this mode, a crystal Oscillator is connected to this pin, and the pin can not be used as an I/O pin.

If PB6 is used as a clock pin, DDB6, PORTB6 and PINB6 will all read 0.

• SCK/PCINT5 – Port B, Bit 5

SCK: Master Clock output, Slave Clock input pin for SPI channel. When the SPI is enabled as a Slave, this pin is configured as an input regardless of the setting of DDB5. When the SPI is enabled as a Master, the data direction of this pin is controlled by DDB5. When the pin is forced by the SPI to be an input, the pull-up can still be controlled by the PORTB5 bit.

PCINT5: Pin Change Interrupt source 5. The PB5 pin can serve as an external interrupt source.

• MISO/PCINT4 – Port B, Bit 4

MISO: Master Data input, Slave Data output pin for SPI channel. When the SPI is enabled as a Master, this pin is configured as an input regardless of the setting of DDB4. When the SPI is enabled as a Slave, the data direction of this pin is controlled by DDB4. When the pin is forced by the SPI to be an input, the pull-up can still be controlled by the PORTB4 bit.

PCINT4: Pin Change Interrupt source 4. The PB4 pin can serve as an external interrupt source.

• MOSI/OC2/PCINT3 – Port B, Bit 3

MOSI: SPI Master Data output, Slave Data input for SPI channel. When the SPI is enabled as a Slave, this pin is configured as an input regardless of the setting of DDB3. When the SPI is enabled as a Master, the data direction of this pin is controlled by DDB3. When the pin is forced by the SPI to be an input, the pull-up can still be controlled by the PORTB3 bit.

OC2, Output Compare Match Output: The PB3 pin can serve as an external output for the

Timer/Counter2 Compare Match. The PB3 pin has to be configured as an output (DDB3 set (one)) to serve this function. The OC2 pin is also the output pin for the PWM mode timer function.

PCINT3: Pin Change Interrupt source 3. The PB3 pin can serve as an external interrupt source.

• SS/OC1B/PCINT2 – Port B, Bit 2

SS: Slave Select input. When the SPI is enabled as a Slave, this pin is configured as an input regardless of the setting of DDB2. As a Slave, the SPI is activated when this pin is driven low. When the SPI is enabled as a Master, the data direction of this pin is controlled by DDB2. When the pin is forced by the SPI to be an input, the pull-up can still be controlled by the PORTB2 bit.

OC1B, Output Compare Match output: The PB2 pin can serve as an external output for the Timer/Counter1 Compare Match B. The PB2 pin has to be configured as an output (DDB2 set (one)) to serve this function. The OC1B pin is also the output pin for the PWM mode timer function.

PCINT2: Pin Change Interrupt source 2. The PB2 pin can serve as an external interrupt source.

• OC1A/PCINT1 – Port B, Bit 1

OC1A, Output Compare Match output: The PB1 pin can serve as an external output for the Timer/Counter1 Compare Match A. The PB1 pin has to be configured as an output (DDB1 set (one)) to serve this function. The OC1A pin is also the output pin for the PWM mode timer function.

PCINT1: Pin Change Interrupt source 1. The PB1 pin can serve as an external interrupt source.

• ICP1/CLKO/PCINT0 – Port B, Bit 0

ICP1, Input Capture Pin: The PB0 pin can act as an Input Capture Pin for Timer/Counter1.

CLKO, Divided System Clock: The divided system clock can be output on the PB0 pin. The divided system clock will be output if the CKOUT Fuse is programmed, regardless of the PORTB0 and DDB0 settings. It will also be output during reset.

PCINT0: Pin Change Interrupt source 0. The PB0 pin can serve as an external interrupt source. The divided system clock will be output if the CKOUT Fuse is programmed, regardless of the PORTB0 and DDB0. The divided system clock can be output on the PB0 pin.

Port C (PC5:0)

Port C is a 7-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The PC5..0 output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running.

PC6/RESET

If the RSTDISBL Fuse is programmed, PC6 is used as an I/O pin. Note that the electrical characteristics of PC6 differ from those of the other pins of Port C.

If the RSTDISBL Fuse is unprogrammed, PC6 is used as a Reset input. A low level on this pin for longer than the minimum pulse length will generate a Reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a Reset. The various special features of Port C are elaborated in table

The alternate pin configuration is as follows:

• RESET/PCINT14 – Port C, Bit 6

RESET, Reset pin: When the RSTDISBL Fuse is programmed, this pin functions as a normal I/O pin, and the part will have to rely on Power-on Reset and Brown-out Reset as its reset sources. When the RSTDISBL Fuse is unprogrammed, the reset circuitry is connected to the pin, and the pin can not be used as an I/O pin. If PC6 is used as a reset pin, DDC6, PORTC6 and PINC6 will all read 0. PCINT14: Pin Change Interrupt source 14. The PC6 pin can serve as an external interrupt source.

• SCL/ADC5/PCINT13 – Port C, Bit 5

SCL, 2-wire Serial Interface Clock: When the TWEN bit in TWCR is set (one) to enable the 2-wire Serial Interface, pin PC5 is disconnected from the port and becomes the Serial Clock I/O pin for the 2-wire Serial Interface. In this mode, there is a spike filter on the pin to suppress spikes shorter than 50 ns on the input signal, and the pin is driven by an open drain driver with slew-rate limitation. PC5 can also be used as ADC input Channel 5. Note that ADC input channel 5 uses digital power. PCINT13: Pin Change Interrupt source 13. The PC5 pin can serve as an external interrupt source. pin PC5 is disconnected from the port and becomes the Serial Clock I/O pin for the 2-wire Serial Interface.

• SDA/ADC4/PCINT12 – Port C, Bit 4

SDA, 2-wire Serial Interface Data: When the TWEN bit in TWCR is set (one) to enable the 2-wire Serial Interface, pin PC4 is disconnected from the port and becomes the Serial Data I/O pin for the 2-wire Serial Interface. In this mode, there is a spike filter on the pin to suppress spikes shorter than 50 ns on the input signal, and the pin is driven by an open drain driver with slewrate limitation. PC4 can also be used as ADC input Channel 4. Note that ADC input channel 4 uses digital power. PCINT12: Pin Change Interrupt source 12. The PC4 pin can serve as an external interrupt source.

• ADC3/PCINT11 – Port C, Bit 3

PC3 can also be used as ADC input Channel 3. Note that ADC input channel 3 uses analog power. PCINT11: Pin Change Interrupt source 11. The PC3 pin can serve as an external interrupt source. In this mode, there is a spike filter on the pin to suppress spikes shorter than 50 ns on the input signal.

• ADC2/PCINT10 – Port C, Bit 2

PC2 can also be used as ADC input Channel 2. Note that ADC input channel 2 uses analog power. PCINT10: Pin Change Interrupt source 10. The PC2 pin can serve as an external interrupt source.

Port D (PD7:0)

Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running.

• AIN1/OC2B/PCINT23 – Port D, Bit 7

AIN1, Analog Comparator Negative Input. Configure the port pin as input with the internal pull-up switched off to avoid the digital port function from interfering with the function of the Analog Comparator. PCINT23: Pin Change Interrupt source 23. The PD7 pin can serve as an external interrupt source. port pin as input with the internal pull-up switched off to avoid the digital port function from interfering with the function of the Analog Comparator . The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running. port pin as input with the internal pull-up switched off to avoid the digital port function from interfering with the function of the Analog Comparator .

• AIN0/OC0A/PCINT22 – Port D, Bit 6

AIN0, Analog Comparator Positive Input. Configure the port pin as input with the internal pull-up switched off to avoid the digital port function from interfering with the function of the Analog Comparator. OC0A, Output Compare Match output: The PD6 pin can serve as an external output for the Timer/Counter0 Compare Match A. The PD6 pin has to be configured as an output (DDD6 set (one)) to serve this function. The OC0A pin is also the output pin for the PWM mode timer function. PCINT22: Pin Change Interrupt source 22. The PD6 pin can serve as an external interrupt source.

• **T1/OC0B/PCINT21 – Port D, Bit 5**

T1, Timer/Counter1 counter source. OC0B, Output Compare Match output: The PD5 pin can serve as an external output for the Timer/Counter0 Compare Match B. The PD5 pin has to be configured as an output (DDD5 set (one)) to serve this function. The OC0B pin is also the output pin for the PWM mode timer function. PCINT21: Pin Change Interrupt source 21. The PD5 pin can serve as an external interrupt source. The PD5 pin has to be configured as an output (DDD5 set (one)) to serve this function.

• **XCK/T0/PCINT20 – Port D, Bit 4**

XCK, USART external clock. T0, Timer/Counter0 counter source. PCINT20: Pin Change Interrupt source 20. The PD4 pin can serve as an external interrupt source.

• **INT1/OC2B/PCINT19 – Port D, Bit 3**

INT1, External Interrupt source 1: The PD3 pin can serve as an external interrupt source. OC2B, Output Compare Match output: The PD3 pin can serve as an external output for the Timer/Counter0 Compare Match B. The PD3 pin has to be configured as an output (DDD3 set (one)) to serve this function. The OC2B pin is also the output pin for the PWM mode timer function. PCINT19: Pin Change Interrupt source 19. The PD3 pin can serve as an external interrupt source.

• **INT0/PCINT18 – Port D, Bit 2**

INT0, External Interrupt source 0: The PD2 pin can serve as an external interrupt source. PCINT18: Pin Change Interrupt source 18. The PD2 pin can serve as an external interrupt source. The OC2B pin is also the output pin for the PWM mode timer function. PCINT19: Pin Change Interrupt source 19. The PD3 pin can serve as an external interrupt source. The PD2 pin can serve as an external interrupt source. PCINT18: Pin Change Interrupt source 18. The PD2 pin can serve as an external interrupt source. The OC2B pin is also the output pin for the PWM mode timer function.

• **TXD/PCINT17 – Port D, Bit 1**

TXD, Transmit Data (Data output pin for the USART). When the USART Transmitter is enabled, this pin is configured as an output regardless of the value of DDD1. PCINT17: Pin Change Interrupt source 17. The PD1 pin can serve as an external interrupt source. The OC2B pin is also the output pin for the PWM mode timer function. PCINT19: Pin Change Interrupt source 19. The PD3 pin can serve as an external interrupt source.

• RXD/PCINT16 – Port D, Bit 0

RXD, Receive Data (Data input pin for the USART). When the USART Receiver is enabled this pin is configured as an input regardless of the value of DDD0. When the USART forces this pin to be an input, the pull-up can still be controlled by the PORTD0 bit. PCINT16: Pin Change Interrupt source 16. The PD0 pin can serve as an external interrupt source.

AVCC

AVCC is the supply voltage pin for the A/D Converter, PC3:0, and ADC7:6. It should be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter. Note that PC6..4 use digital supply voltage, VCC.

AREF

AREF is the analog reference pin for the A/D Converter.

ADC7:6 (TQFP and QFN/MLF Package Only)

In the TQFP and QFN/MLF package, ADC7:6 serve as analog inputs to the A/D converter. These pins are powered from the analog supply and serve as 10-bit ADC channels.

3.2 REGULATED POWER SUPPLY

In mains-supplied electronic systems the AC input voltage must be converted into a DC voltage with the right value and degree of stabilization. The common DC voltages that are required to power up the devices are generally in the range of 3 VDC to 30 VDC. Typically the fixed types of DC voltages are 5V, 9V, 12V, 15V and 18V DC. the AC input voltage must be converted into a DC voltage with the right value and degree of stabilization.

Power supply Modules:

- Step down transformer
- Bridge rectifier with filter

Regulated Power Supply Block diagram:



Fig 3.3: Regulated power supply

3.2.1 TRANSFORMER

Transformers convert AC electricity from one voltage to another with little loss of power. Transformers work only with AC and this is one of the reasons why mains electricity is AC. Step-up transformers increase voltage, step-down transformers reduce voltage.

A step down power transformer is used to step down the AC voltage from the line voltage of 110 VAC or 220 VAC i.e, it converts higher voltage at the input side(V_{in}) to a lower voltage at the output (V_{out}).

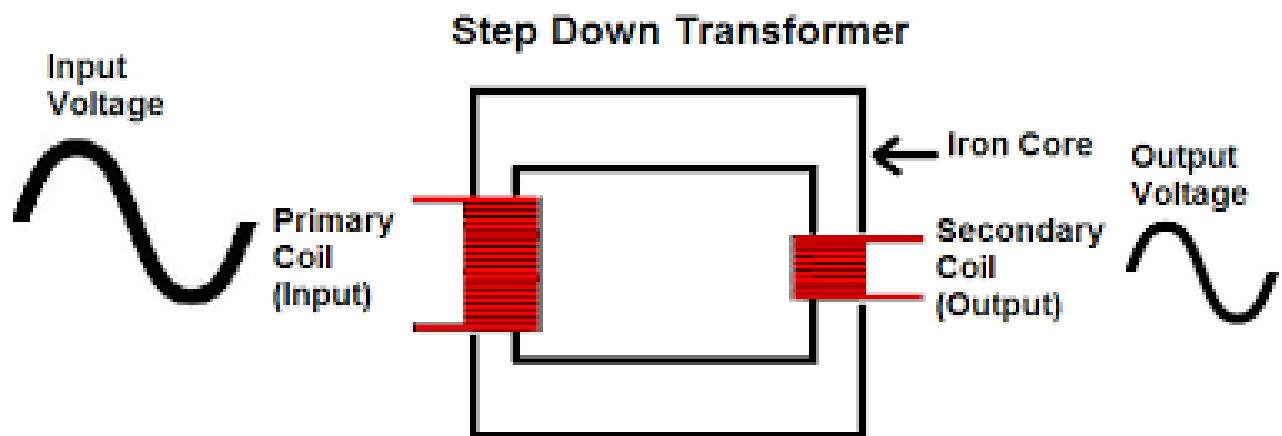


Fig 3.4: Transformer

3.2.2 RECTIFIER

There are several ways of connecting diodes to make a rectifier to convert AC to DC. The bridge rectifier is the most important and it produces full-wave varying DC. A bridge rectifier makes use of four diodes in a bridge arrangement to achieve full-wave rectification. A rectifier is an electronic device that converts alternating current (AC) into direct current (DC) using one or more P-N junction diodes.

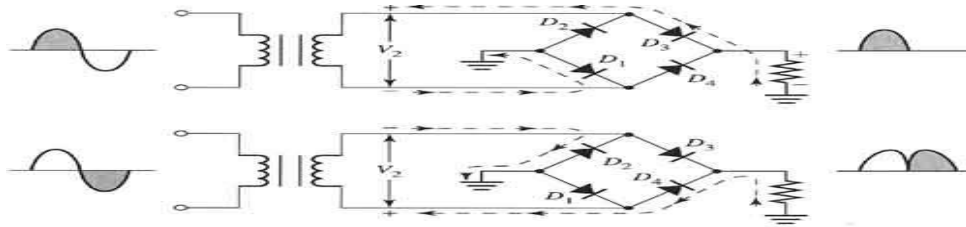


Fig 3.5: Rectifier

3.2.3 FILTER

Filtering is performed by a large value electrolytic capacitor connected across the DC supply to act as a reservoir, supplying current to the output when the varying DC voltage from the rectifier is falling. The diagram shows the unfiltered varying DC (dotted line) and the filtered DC (solid line). The capacitor charges quickly near the peak of the varying DC, and then discharges as it supplies current to the output. Typically 1000 μF capacitor is used.

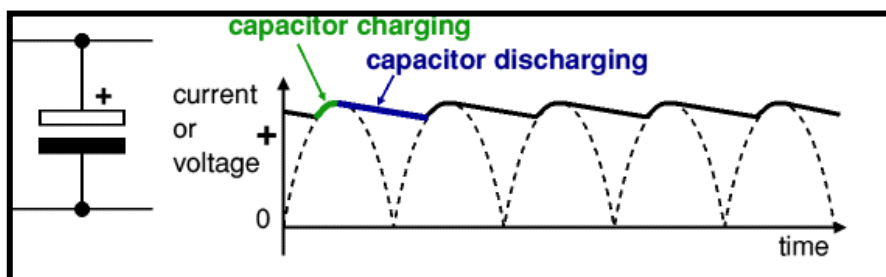


Fig 3.6: Filter

3.2.4 REGULATOR

This is a simple DC regulated supply project using 7805 voltage regulator to obtain a variable DC voltage range from 5V to 15V.

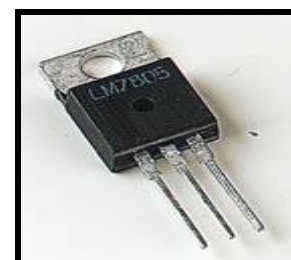
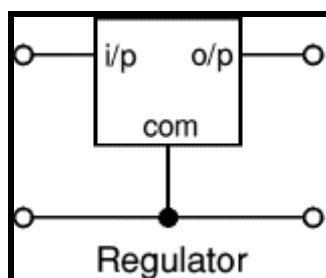


Fig 3.7: Pin out of 7804

Pin out of the 7805 regulator IC.

1. Unregulated voltage in

2. Ground

3. Regulated voltage out

Circuit Diagram of R. P. S

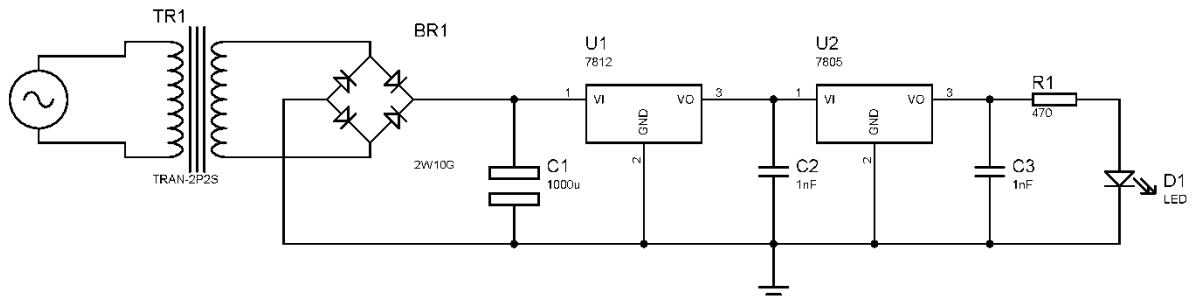


Fig 3.8: RPS

3.6 Light Emitting Diode (LED)

LEDs are everywhere – in our phones, in our cars, and even in our homes. Whenever an electronic device lights up, there is a good chance that an LED is behind it. LEDs are like tiny lightbulbs. Low energy consumption, small size, rapid switching and long lifespan makes them ideal for mobile devices and other low-power application.

LED stands for Light Emitting Diode. They are a special type of diode that convert electrical energy into light. They have very similar electrical characteristics to a normal PN junction diode. That's why the symbol of LED is similar to the normal PN junction diode except that it contains arrows pointing away from the diode indicating that light is being emitted by the diode.

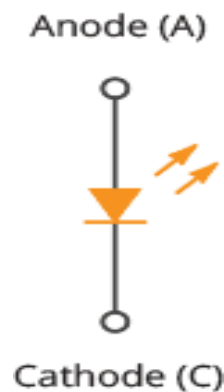


Fig 3.9: Symbol of LED

3.6.1 LED Construction

LEDs are so common, they come in a huge variety of shapes, sizes and colors. The LEDs you are most likely to use are the standard through hole LEDs with two legs. Following figure shows the parts of The construction of an LED is very different from an ordinary diode. The PN junction of an LED is surrounded by a transparent, rigid plastic epoxy resin shell. The shell is constructed in such a way that photons of light emitted by the junction are focused upward through the domed top of the LED.

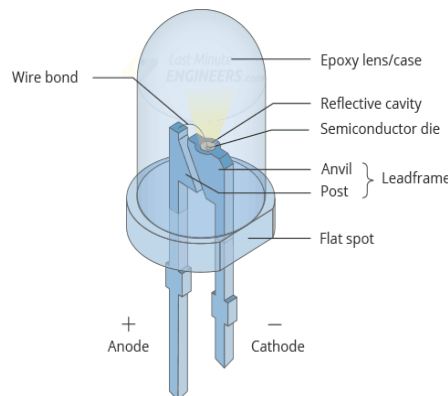


Fig 3.10: LED

Just as in an ordinary diode, the positive side of the LED is called the Anode, while the negative side of the LED is called the Cathode. The cathode is usually indicated by having a shorter lead than the anode. Not only this, the outside of the plastic case typically has a flat spot or notch which can also indicate the cathode side of the LED. Not all LEDs are hemispherical in shape, some are rectangular while some are cylindrical, but they are mostly constructed in the same way.

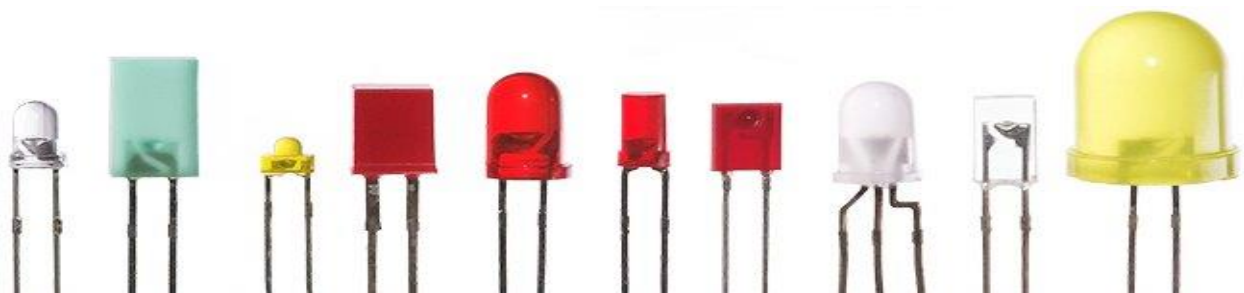


Fig 3.11: Different Shapes of LED

3.6.2 LED Working

Like an ordinary diode, the LED operates only in forward bias condition. When the LED is forward biased, the free electrons cross the PN junction and recombine with holes. Since these electrons fall from a higher to a lower energy level, they radiate energy in the

form of photons (light). In ordinary diodes, this energy is radiated as heat while in an LED, energy is radiated as This effect is called Electroluminescence.

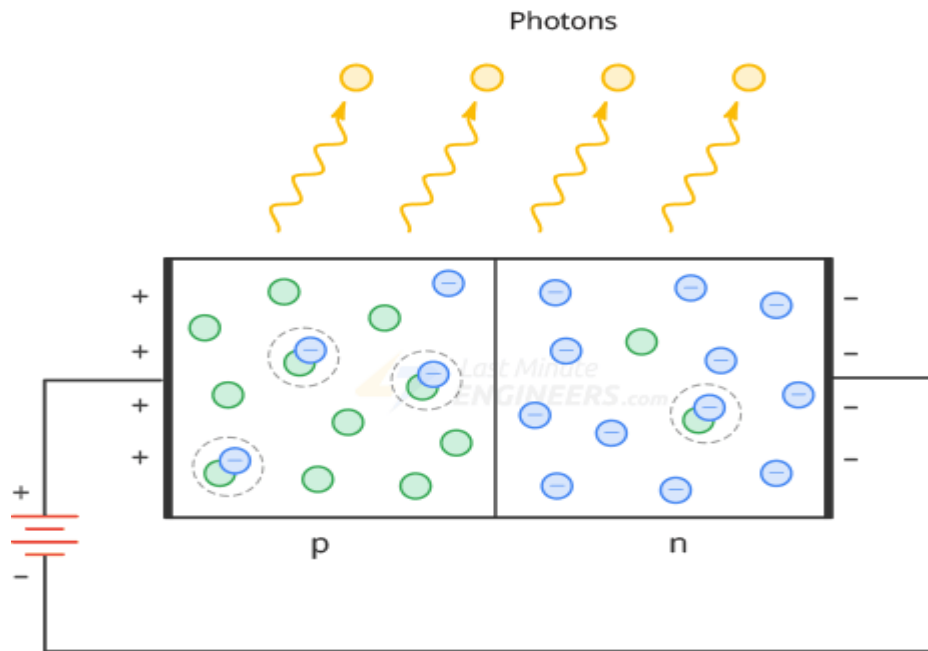


Fig3.12 Working of LED

3.4 Buzzer

There are many ways to communicate between the user and a product. One of the best ways is audio communication using a buzzer IC. So during the design process, understanding some technologies with configurations is very helpful. So, this article discusses an overview of an audio signaling device like a beeper or a buzzer and its working with applications.

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.

There are two kinds of buzzers commonly available like simple and readymade. Once a simple type is power-driven then it will generate a beep sound continuously. A readymade type looks heavier & generates a Beep. Beep. Beep. This sound is because of the internal oscillating circuit within it. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. So during the design process, understanding some technologies with configurations is very helpful.



Fig 3.13 buzzer terminals

The pin configuration of the buzzer is shown below. It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-' symbol or short terminal and it is connected to the GND terminal.

A buzzer is available in different types which include the following.

- Piezoelectric
- Electromagnetic
- Mechanical
- Electromechanical
- Magnetic

3.4.1 Working Principle

The working principle of a buzzer depends on the theory that, once the voltage is given across a piezoelectric material, then a pressure difference is produced.

A piezo type includes piezo crystals among two conductors.

Once a potential disparity is given across these crystals, then they thrust one conductor & drag the additional conductor through their internal property. So this continuous action will produce a sharp sound signal.

Mounting Configurations

The mounting configurations of buzzers include the following.

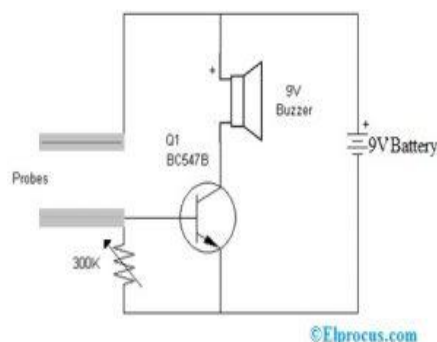
- Panel Mount

- Wire Leads
- Screw Terminals
- Through Hole
- Spring Contact
- Surface Mount

A buzzer is an efficient component to include the features of sound in our system or project. It is an extremely small & solid two-pin device thus it can be simply utilized on breadboard or PCB. So in most applications, this component is widely used. There are two kinds of buzzers commonly available like simple and readymade. Once a simple type is power-driven then it will generate a beep sound continuously. A readymade type looks heavier & generates a Beep. Beep. Beep. This sound is because of the internal oscillating circuit within it. This buzzer uses a DC power supply that ranges from 4V – 9V. To operate this, a 9V battery is used but it is suggested to utilize a regulated +5V/+6V DC supply.

Buzzer Circuit Diagram

The circuit diagram of the water level indicator using the buzzer is shown below. This circuit is used to sense or detect the water level within the tank or washing machine or pool, etc. This circuit is very simple to design using few components such as a transistor, buzzer, 300K variable resistor, and power supply or 9V battery.



3.14 Water Level Circuit using Buzzer

Once the two probes of the circuit are placed in the tank, it detects the level of water. Once the water level exceeds the fixed level, then it generates a beep sound through a buzzer connected to the circuit. This circuit uses a BC547B NPN transistor however we can also use any general-purpose transistor instead of using 2N3904/2N2222.

This water level sensor circuit working is very simple and the transistor used within the circuit works as a switch. Once the two probes notice the water level within the tank, then the transistor turns ON & the voltage begins flowing throughout the transistor to trigger the buzzer.

3.4.2 Specifications

The specifications of the buzzer include the following.

- Color is black
- The frequency range is 3,300Hz
- Operating Temperature ranges from – 20° C to +60°C
- Operating voltage ranges from 3V to 24V DC
- The sound pressure level is 85dBA or 10cm
- The supply current is below 15mA

3.5 GSM

3.5.1 GSM OVERVIEW

Global System for Mobile Communications (GSM) is a standard developed by the European Telecommunications Standards Institute (ETSI) to describe protocols for second generation (2G) digital cellular networks use by mobile phones. It is the default global standard for mobile communication with over 90% global market and is available in over 219 countries and territories. This ubiquity means that subscribers can use their phones throughout the world, enabled by international roaming arrangements

between mobile network operators. GSM differs from its predecessor technologies in that both signaling and speech channels are digital. This also facilitates the wide-spread implementation of data communication applications into the system. The GSM standard has been an advantage to both consumers, who may benefit from the ability to roam and switch carriers without replacing phones, and also to network operators, who can choose equipment from many GSM equipment vendors. GSM also pioneered low-cost implementation of the short message service (SMS), also called text messaging, which has since been supported on other mobile phone standards as well.

3.5.2 GSM SPECIFICATIONS

Frequency: 900 MHz or 1800 MHz (Some countries in the Americas including Canada and the United States use the 850 MHz and 1900 MHz bands, 400 and 450 MHz frequency bands are assigned in some countries, notably Scandinavia)

Modulation: Modulation is a form of change process where we change the input information into a suitable format for the transmission medium. We also changed the information by demodulating the signal at the receiving end. The GSM uses Gaussian Minimum Shift Keying (GMSK) modulation method.

Access Methods: Because radio spectrum is a limited resource shared by all users, a method must be devised to divide up the bandwidth among as many users as possible. GSM chose a combination of TDMA/FDMA as its method. The FDMA part involves the division by frequency of the total 25 MHz bandwidth into 124 carrier frequencies of 200 kHz bandwidth. One or more carrier frequencies are then assigned to each BS. Each of these carrier frequencies is then divided in time, using a TDMA scheme, into eight time slots. One time slot is used for transmission by the mobile and one for reception. They are separated in time so that the mobile unit does not receive and transmit at the same time.

Transmission Rate: The total symbol rate for GSM at 1 bit per symbol in GMSK produces 270.833 K symbols/second. The gross transmission rate of the time slot is 22.8 Kbps. . Each of these carrier frequencies is then divided in time, using a TDMA scheme, into eight time slots.

Frequency Band: The uplink frequency range specified for GSM is 933 - 960 MHz (basic 900 MHz band only). The downlink frequency band 890 - 915 MHz (basic 900 MHz band only).

Channel Spacing: This indicates separation between adjacent carrier frequencies. In GSM, this is 200 kHz.

Speech Coding: GSM uses linear predictive coding (LPC). The purpose of LPC is to reduce the bit rate. The LPC provides parameters for a filter that mimics the vocal tract. The signal passes through this filter, leaving behind a residual signal. Speech is encoded at 13 kbps.

Duplex Distance: The duplex distance is 80 MHz Duplex distance is the distance between the uplink and downlink frequencies. A channel has two frequencies, 80 MHz apart.

3.5.3 GSM MODEM

GSM modem is a specialized type of modem which accepts a SIM card, and operates over a subscription to a mobile operator, just like a mobile phone. From the mobile operator perspective, a GSM modem looks just like a mobile phone. A wireless modem behaves like a dial-up modem. The main difference between them is that a dial-up modem sends and receives data through a fixed telephone line while a wireless modem sends and receives data through radio waves.

A GSM modem can be an external device or a PC Card / PCMCIA Card. Typically, an external GSM modem is connected to a computer through a serial cable or a USB cable. A GSM modem in the form of a PC Card / PCMCIA Card is designed for use with a laptop computer. It should be inserted into one of the PC Card / PCMCIA Card slots of a laptop computer. Like a GSM mobile phone, a GSM modem requires a SIM card from a wireless carrier in order to operate. Both GSM modems and dial-up modems support a common set of standard AT commands. You can use a GSM modem just like a dial-up modem. A GSM modem in the form of a PC Card / PCMCIA Card is designed for use with a laptop computer.

A wireless modem behaves like a dial-up modem. The main difference between them is that a dial-up modem sends and receives data through a fixed telephone line while a wireless modem sends and receives data through radio waves. It should be inserted into one of the PC Card / PCMCIA Card slots of a laptop computer.



Fig.3.15 GSM modem

3.5.4 ARCHITECTURE OF THE GSM NETWORK

In a GSM network, the user terminal is called a mobile station. A mobile station is made up of a SIM (Subscriber Identity Module) card allowing the user to be uniquely identified and a mobile terminal.

The terminals (devices) are identified by a unique 15-digit identification number called IMEI (International Mobile Equipment Identity). Each SIM card also has a unique (and secret) identification number called IMSI (International Mobile Subscriber Identity). This code can be protected using a 4-digit key called a PIN code.

The SIM card therefore allows each user to be identified independently of the terminal used during communication with a base station. Communications occur through a radio link (air interface) between a mobile station and a base station.

All the base stations of a cellular network are connected to a base station controller (BSC) which is responsible for managing distribution of the resources. The system consisting of the base station controller and its connected base stations is called the Base Station Subsystem (BSS). Finally, the base station controllers are themselves physically connected to

the Mobile Switching Centre (MSC), managed by the telephone network operator, which connects them to the public telephone network and the Internet.

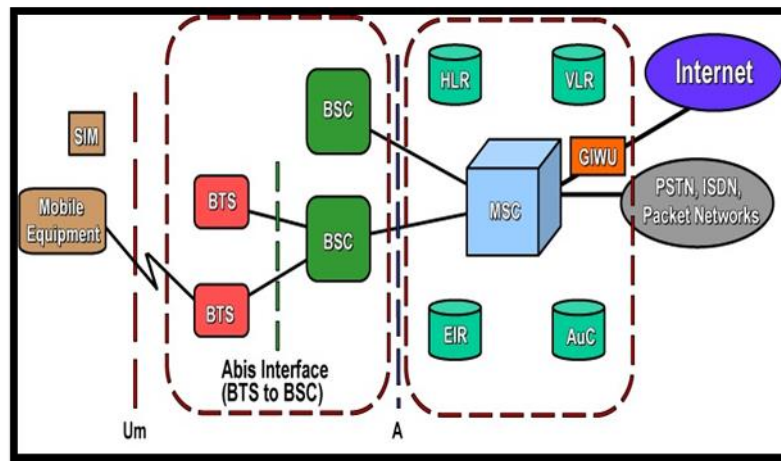


Fig.3.16 GSM Architecture

The MSC belongs to a Network Station Subsystem (NSS), which is responsible for managing user identities, their location and establishment of communications with other subscribers. The MSC is generally connected to databases that provide additional functions:

- The Home Location Register (HLR) is a database containing information (geographic position, administrative information etc.) of the subscribers registered in the area of the switch (MSC).
- The Visitor Location Register (VLR) is a database containing information of users other than the local subscribers. The VLR retrieves the data of a new user from the HLR of the user's subscriber zone. The data is maintained as long as the user is in the zone and is deleted when the user leaves or after a long period of inactivity (terminal off).
- The Equipment Identify Register (EIR) is a database listing the mobile terminals.
- The Authentication Centre (AUC) is responsible for verifying user identities.
- The cellular network formed in this way is designed to support mobility via management of handovers (movements from one cell to another).

3.5.5 INTRODUCTION TO AT COMMANDS

AT commands are instructions used to control a modem. AT is the abbreviation of Attention. Every command line starts with "AT" or "at". That's the reason, modem commands

are called AT commands. Many of the commands that are used to control wired dial-up modems. The system consisting of the base station controller and its connected base stations is called the Base Station

COMMANDS	DESCRIPTION
A/	Re-Issue Last AT Command
ATA	Answer Incoming Call
ATH	Disconnect Existing Connection
AT+CMGD	Delete SMS Message
AT+CMGF	Select Message Format
AT+CMGL	List of SMS Message
AT+CMGR	Read the SMS Message
AT+CMGS	Send SMS Message
AT+CNMI	How Modem Will Response When A SMS is Recieved
AT+CPIN?	To Check SIM is Ready
AT+CREG?	Network Register

Tabel.3.1 commands table

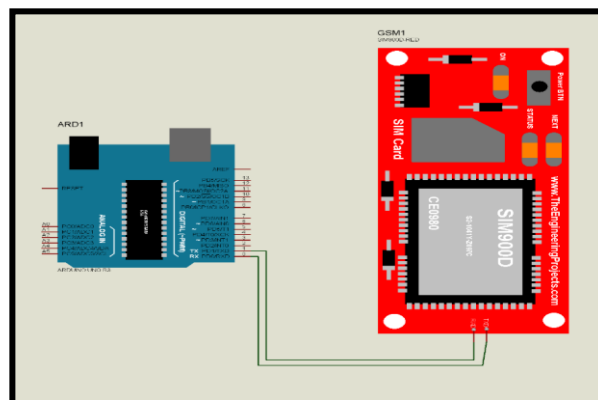


Fig:3.17 Interfacing GSM modem with ARDUINO UNO

3.6 Water level sensors

Water level sensors are essential components in flood alert and rescue systems, designed to detect rising water levels in rivers, reservoirs, urban drainage, or flood-prone areas. These systems commonly use sensors like ultrasonic and radar types due to their high accuracy and non-contact nature, making them resilient against debris and environmental conditions such as heavy rain and fog. Ultrasonic sensors emit sound waves to measure distance from the water surface, while radar sensors use radar waves, providing reliable readings even in harsh weather. Pressure sensors are also widely used in submerged applications, measuring the hydrostatic pressure of water to determine level changes,

especially suitable for deep water bodies and turbulent conditions. Float switch sensors, though simpler, can be effective in small-scale applications, like storm drains, triggering alerts when water reaches a specific level. For effective flood alerting, these sensors often integrate with real-time data transmission systems, sending instant updates to central monitoring stations through cellular, satellite, or IoT networks. They can activate alarms or notifications (via SMS or mobile apps), enabling rapid communication with local authorities and residents. Reliable power sources, such as solar with battery backups, are crucial in ensuring uninterrupted operation during power outages. In many setups, redundant sensors placed at various points provide a comprehensive view of rising water across large areas, improving early detection and alert accuracy. By continuously gathering data, these systems can also support predictive flood models, aiding responders in anticipating flood paths and potential impacts, ultimately reducing risks and enhancing emergency preparedness.

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Fig:3.18 water level sensor

3.6.1 working of water level sensor

Water level sensors in flood alerting and rescue systems work by continuously monitoring changes in water levels and sending real-time data to alert authorities and communities of potential flood risks. Sensors like ultrasonic and radar types operate by emitting waves—either sound or radar—toward the water surface and measuring the time it takes for the waves to reflect back.

This allows them to accurately detect rising water even in turbulent conditions. Sensors like ultrasonic and radar types operate by emitting waves—either sound or radar—toward the water surface and measuring the time it takes for the waves to reflect back. This allows them to accurately detect rising water even in turbulent conditions. Pressure sensors, commonly used in submerged environments, measure water levels based on the hydrostatic pressure exerted by the water column above the sensor, making them ideal for monitoring rivers and reservoirs. Whether you're looking for a water level sensor for irrigation systems, drinking water systems, or industrial applications, there are many options available on the market.

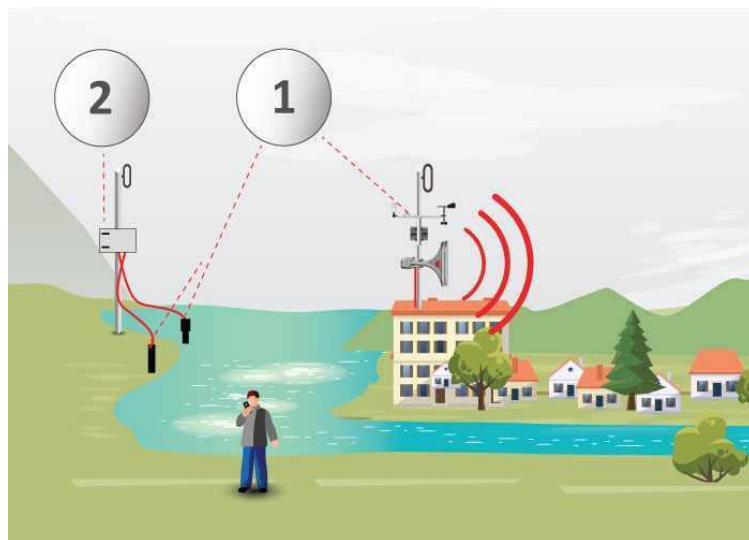


Fig:3.19 working of water level sensor

These alerts can activate sirens, send SMS notifications, or even launch evacuation orders. To ensure continuous operation during floods, especially during power outages, sensors are often equipped with solar-powered batteries. Distributed sensors across multiple locations offer a broader, more accurate picture of water levels, allowing authorities to make informed decisions quickly. By integrating data analytics, some systems also support predictive modeling, helping emergency responders anticipate flood patterns and prioritize rescue efforts, ultimately reducing damage and enhancing community safety. Water level sensors are essential tools for monitoring water levels in various applications. They provide accurate and reliable readings and help to ensure that there is an efficient use of water and avoid water shortages or waste. Whether you're looking for a water level sensor for irrigation systems, drinking water systems, or industrial applications, there are many options available on the market.

Consider your specific needs and choose the best option for your application. Water level sensors are used in flood warning systems to monitor the water level in rivers and lakes. This helps to provide early warnings of potential flooding, allowing communities to prepare and evacuate if necessary. By integrating data analytics, some systems also support predictive modeling, helping emergency responders anticipate flood patterns and prioritize rescue efforts, ultimately reducing damage and enhancing community safety.

Water level sensors are essential tools for monitoring water levels in various applications. They provide accurate and reliable readings and help to ensure that there is an efficient use of water and avoid water shortages or waste. Whether you're looking for a water level sensor for irrigation systems, drinking water systems, or industrial applications, there are many options available on the market. Consider your specific needs and choose the best option for your application. By integrating data analytics, some systems also support predictive modeling, helping emergency responders anticipate flood patterns and prioritize rescue efforts, ultimately reducing damage and enhancing community safety. Water level sensors are essential tools for monitoring water levels in various applications. They provide accurate and reliable readings and help to ensure that there is an efficient use of water and avoid water shortages or waste. Whether you're looking for a water level sensor for irrigation systems, drinking water systems, or industrial applications, there are many options available on the market.

CHAPTER -4

SOFTWARE DESCRIPTION

4.1 Programming Arduino

Once Arduino IDE is installed on the computer, connect the board with computer using USB cable. Now open the Arduino IDE and choose the correct board by selecting Tools>Boards>Arduino/Genuine Uno, and choose the correct Port by selecting Tools>Port. NODE MCU is programmed using Arduino programming language based on Wiring. To get it started with NODE MCU board and blink the built-in LED, load the example code by selecting Files>Examples>Basics>Blink. Once the example code (also shown below) is loaded into your IDE, click on the ‘upload’ button given on the top bar. Once the upload is finished, you should see the Arduino’s built-in LED blinking. Below is the example code for blinking:

4.2 ARDUINO – INSTALLATION

After learning about the main parts of the NODE MCU board, we are ready to learn how to set up the Arduino IDE. Once we learn this, we will be ready to upload our program on the Arduino board. In this section, we will learn in easy steps, how to set up the Arduino Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: About Arduino board First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use NODE MCU, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image. In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image in case you use NODE MCU, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug). In case you use NODE MCU, Arduino Duemilanove, Nano, Arduino Mega. the kind you would connect to a USB printer as shown in the following image. In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image in case you use NODE MCU. In case you use NODE MCU, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.



Fig4.1 USB Cable

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image



Fig 4.2 A TO MINI-B CABLE

Step 2: Download Arduino IDE Software

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

Step 3: Power up your board

The NODE MCU, Mega, Duemilanove, Arduino, and nano technology automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection.

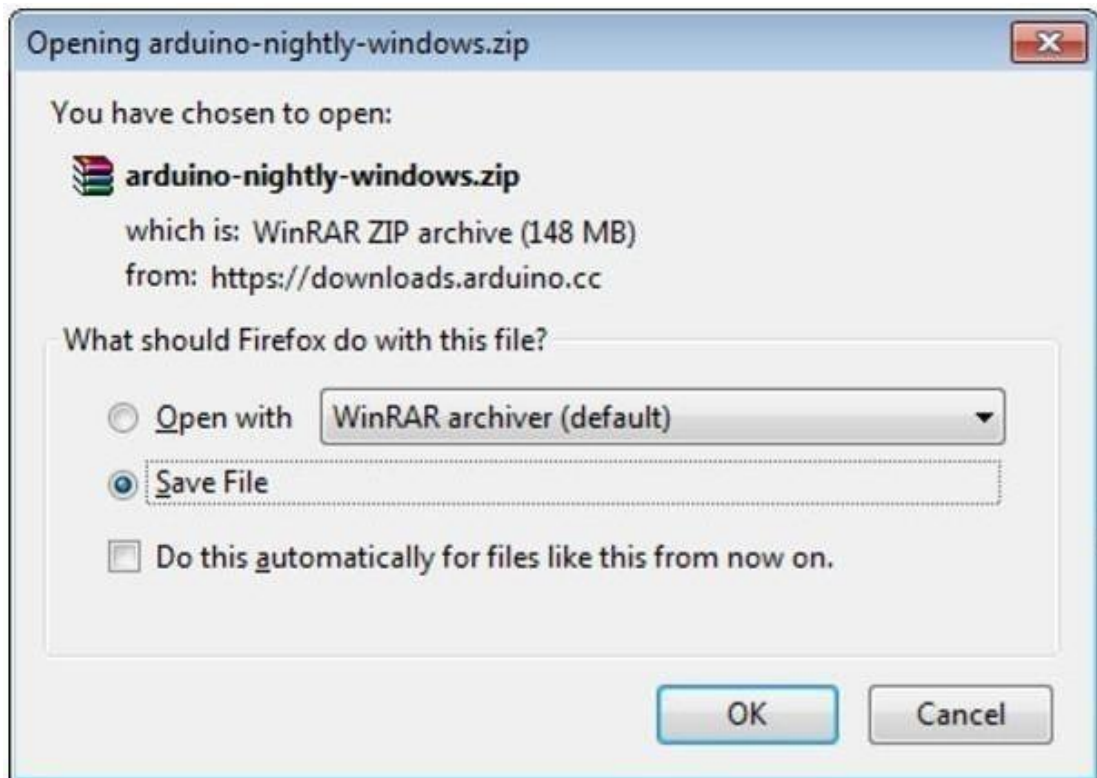


Fig 4.3 Opening Arduino

The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4: Launch Arduino IDE

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-Click the icon to start the IDE. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow. Double-Click the icon to start the IDE. Check that it is on the two pins closest to the USB port. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-Click the icon to start the IDE. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

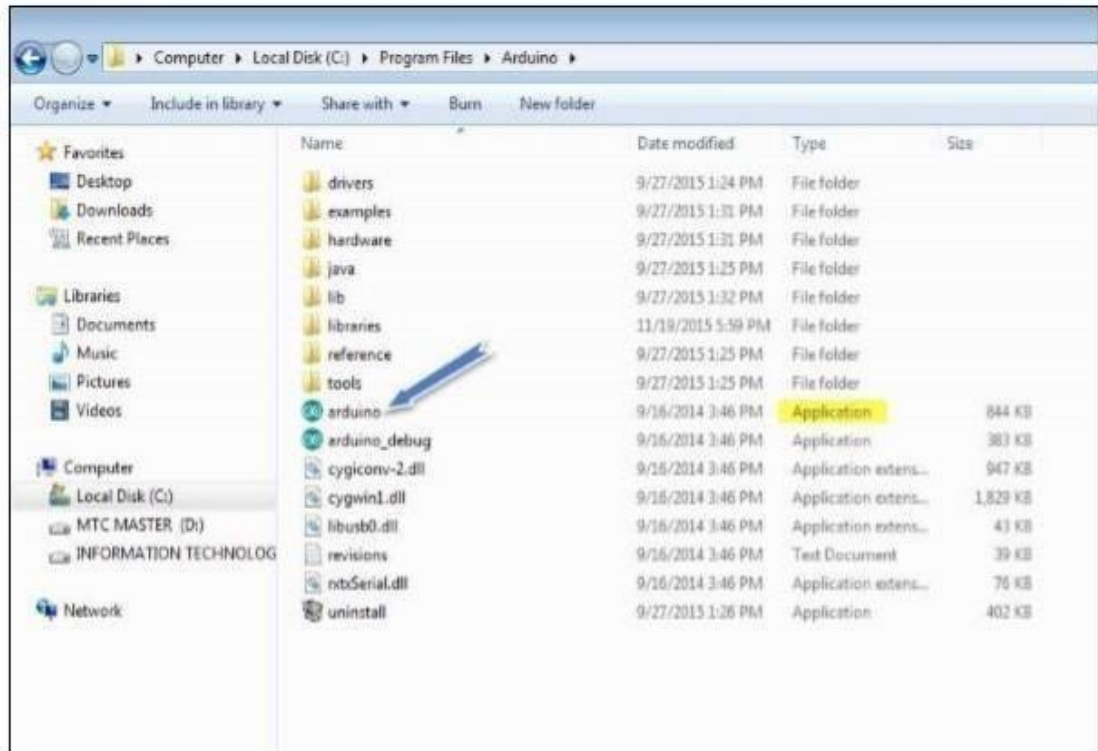


Fig 4.4 Launching Arduino IDE

Step 5: Open your first project

Once the software starts, you have two options:

- ☐ Create a new project.
- ☐ Open an existing project example.

To create a new project, select File --> New

To open an existing project example, select File -> Example ->

Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list.

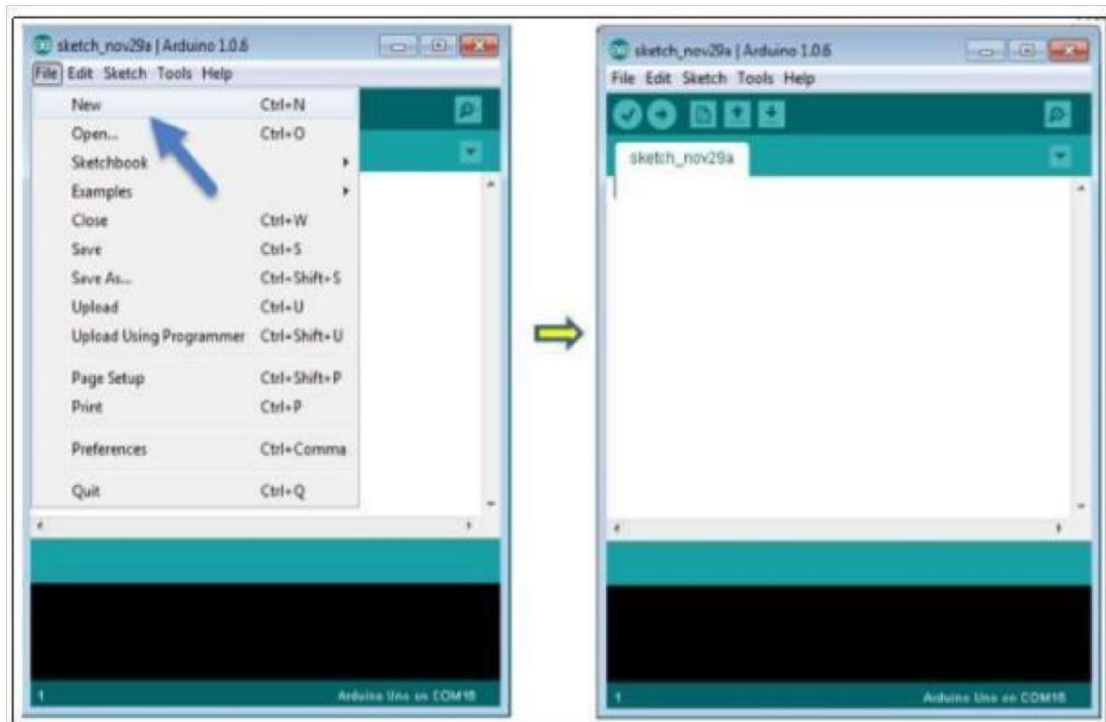


Fig 4.5 Arduino Software Opening New File

Basics->Blink.

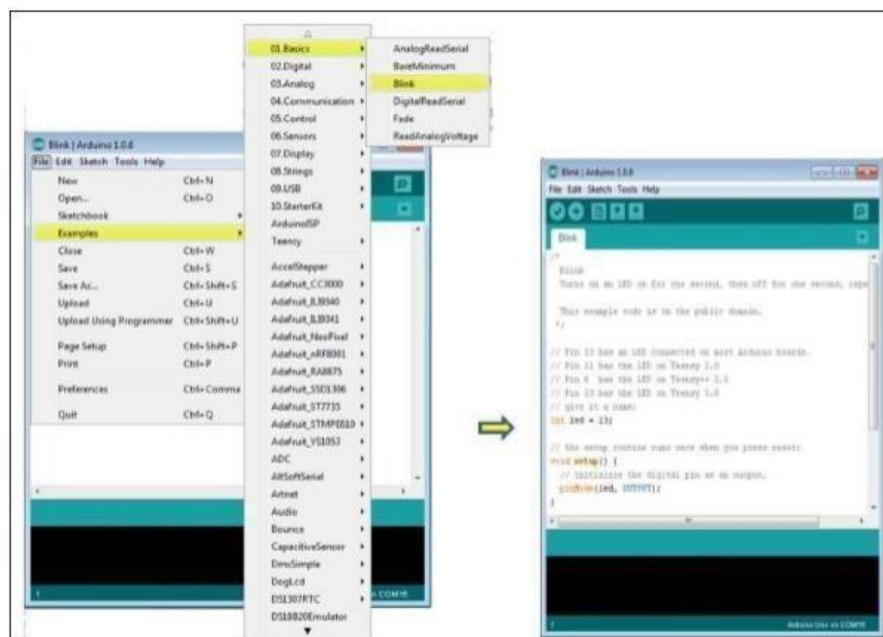


Fig 4.6 Selecting File Type

Step 6: Select your Arduino board

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

Go to Tools -> Board and select your board.

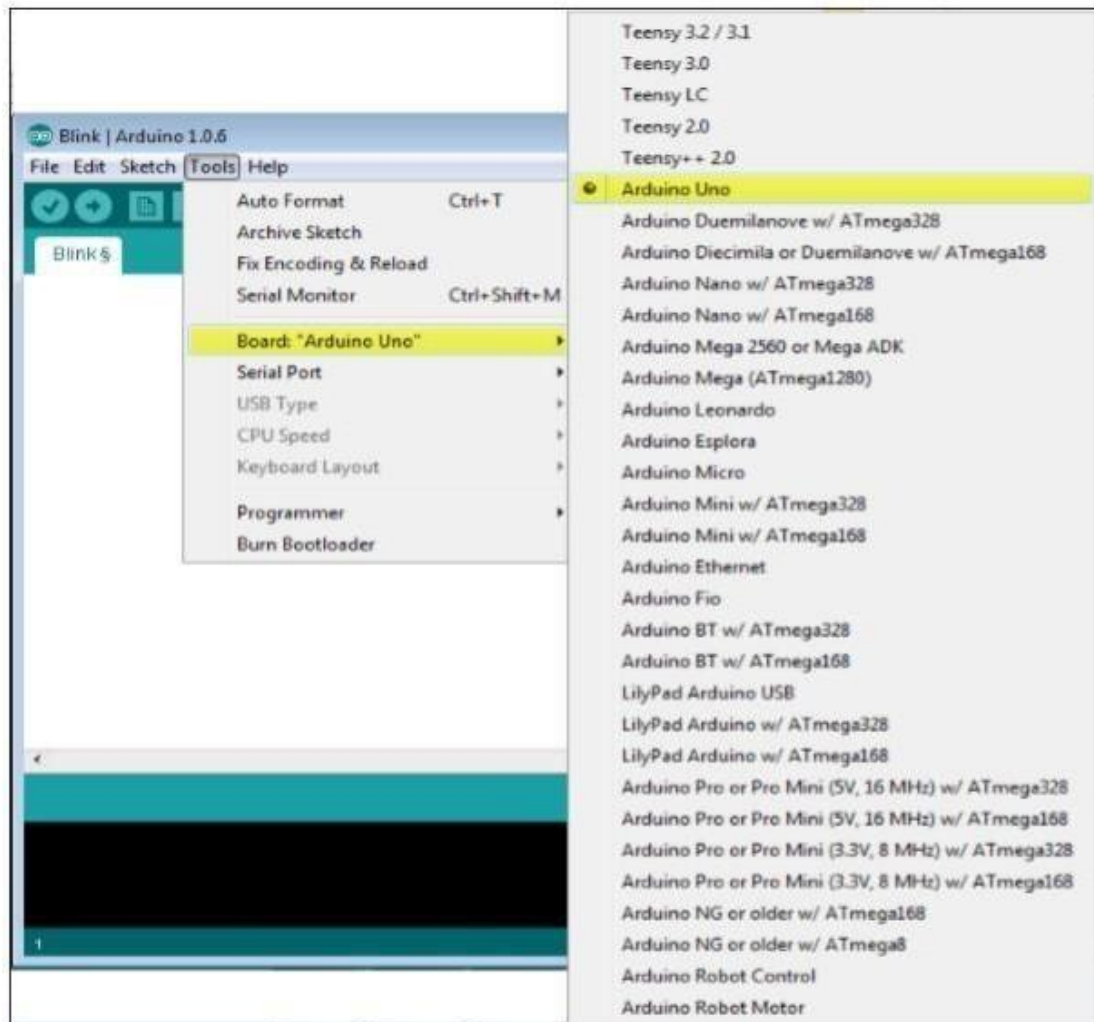


Fig 4.7 Selecting Board Type

Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are using.

Step 7: Select your serial port

Select the serial device of the Arduino board. Go to Tools -> Serial Port menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.

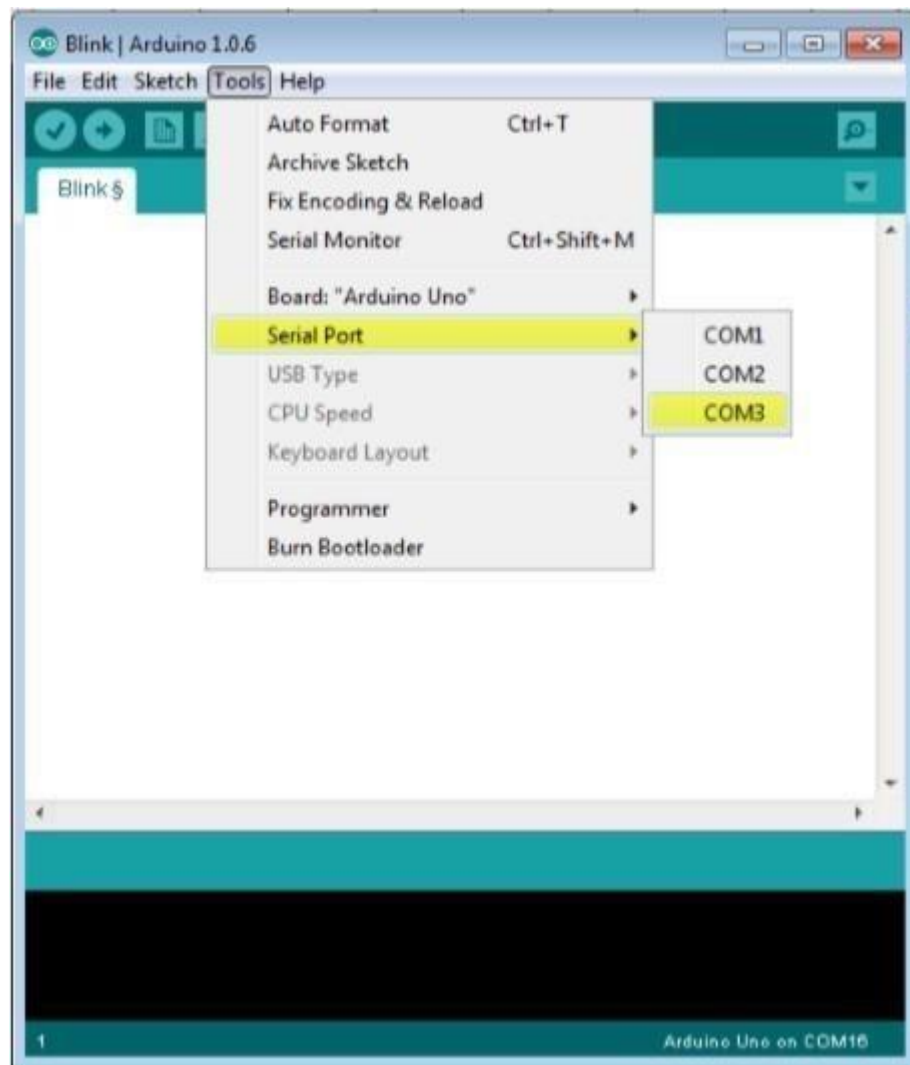


Fig 4.8 Selecting Serial Port

Step 8: Upload the program to your board Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar

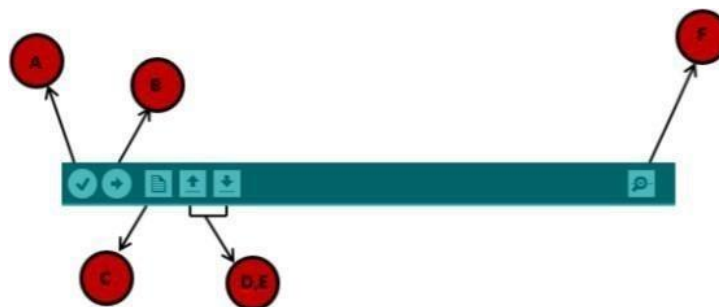


Fig 4.9 Uploading Program To Board

- A. Used to check if there is any compilation error.
- B. Used to upload a program to the Arduino board.
- C. Shortcut used to create a new sketch.
- D. Used to directly open one of the example sketch.
- E. Used to save your sketch.
- F. Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

Note: If you have an Arduino Mini, NG, or other board, you need to press the reset button physically on the board, immediately before clicking the upload button on the Arduino Software.

4.3 ARDUINO – PROGRAM STRUCTURE

We will study in depth, the Arduino program structure and we will learn more new terminologies used in the Arduino world. The Arduino software is opensource. The source code for the Java environment is released under the GPL and the C/C++ microcontroller libraries are under the LGPL. Sketch: The first new terminology is the Arduino program called “sketch”. Structure Arduino programs can be divided in three main parts: Structure, Values (variables and constants), and Functions. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Structure Arduino programs can be divided in three main parts: Structure, Values (variables and constants), and Functions. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Let us start with the Structure. Software structure consist of two main Function:

□ Setup() function

□ Loop() function

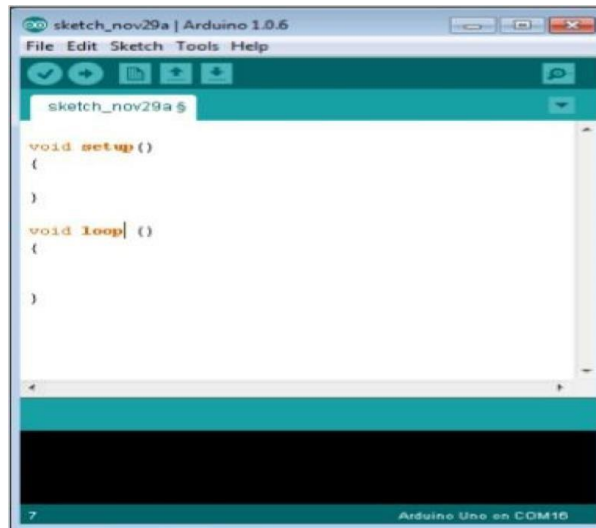


Fig 4.10 Program Structure

```
Void setup ( )  
{  
  
}
```

PURPOSE: The setup () function is called when a sketch starts. Use it to initialize the variables, pin modes, start using libraries, etc.

Arduino board.

INPUT:

OUTPUT:

RETURN:

```
Void Loop ( )  
{  
  
}
```

PURPOSE: After creating a setup () function, which initializes and set the initial values, the loop () function does precisely what its name suggests, and loops consecutively, allowing your program to change and respond.

INPUT:

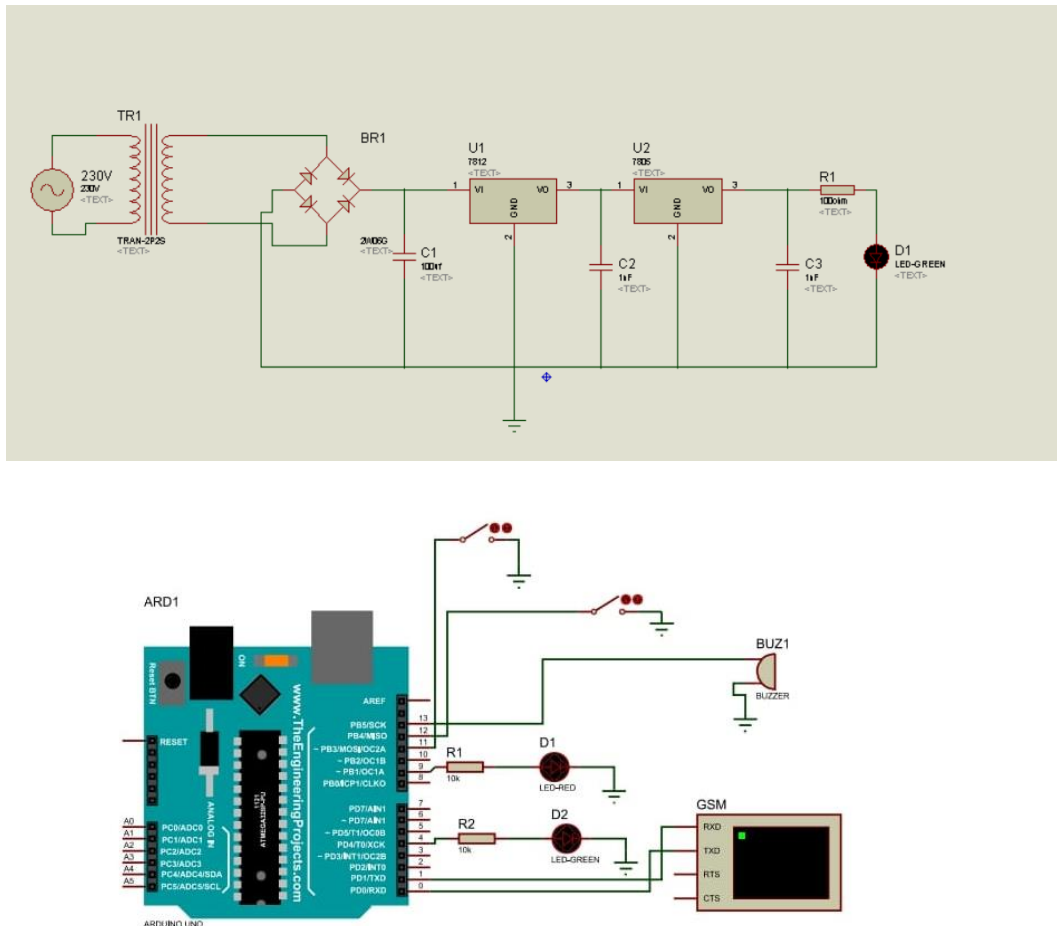
OUTPUT:

RETURN:

CHAPTER-5

DESIGN & IMPLEMENTATION

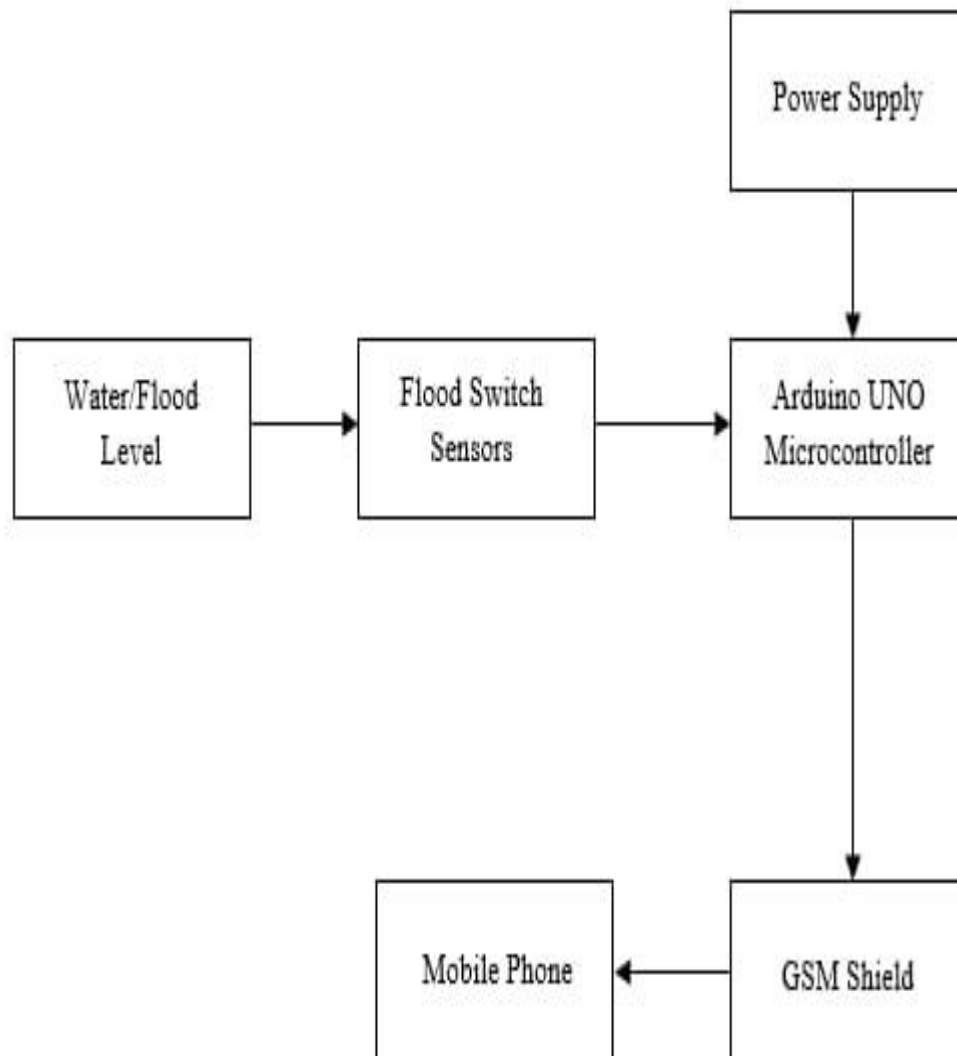
5.1 Block diagram overview



A flood alerting and rescue system using GSM and Arduino works by monitoring water levels with sensors connected to an Arduino microcontroller, which acts as the system's central processor. Water level sensors—such as ultrasonic, float, or pressure sensors—are placed in vulnerable areas, like rivers, drainage systems, or reservoirs, to detect rising water. When the sensors detect that water has reached a dangerous level, the Arduino processes this information and sends an alert via a GSM module. The GSM module, connected to the Arduino, enables the system to send SMS messages or make calls to pre-programmed contacts, such as local authorities or residents, notifying them of impending flood risks. The system can be programmed to send updates periodically,

allowing continuous monitoring and rapid responses to changing conditions. Since the setup is relatively low-cost and power-efficient.

5.2 Flow chart



CHAPTER -6

ADVANTAGES & APPLICATIONS

6.1 ADVANTAGES:

1. **Early Warning:** These systems provide timely alerts to residents and authorities about impending floods, allowing for proactive measures to be taken. Early warnings can significantly reduce the risk of casualties and injuries.
2. **Improved Preparedness:** By disseminating information about potential flooding, these systems enable communities to prepare better. Residents can secure their properties, gather emergency supplies, and establish evacuation plans, reducing panic during an actual event.
3. **Efficient Resource Allocation:** Coordinated flood response efforts ensure that emergency services can deploy resources where they are most needed. This includes identifying safe evacuation routes and directing rescue teams to affected areas promptly.
4. **Data-Driven Decision-Making:** Flood alerting systems rely on real-time data and predictive modeling, which empower local authorities to make informed decisions regarding evacuations and resource distribution.

6.2 APPLICATIONS:

1. **Emergency Response Coordination:** During a flood event, these systems facilitate the coordination of emergency services, ensuring that rescue teams are dispatched efficiently to affected areas and that evacuation routes are clear.
2. **Risk Assessment and Mapping:** Flood alerting systems often incorporate GIS and mapping tools to assess flood risks and create detailed flood hazard maps. This information is valuable for urban planning and community preparedness.
3. **Public Education and Awareness:** Many systems include outreach programs to educate the public about flood risks, preparedness measures, and evacuation protocols, fostering a culture of resilience. This information is valuable for urban planning and community preparedness.

CHAPTER -7

RESULTS

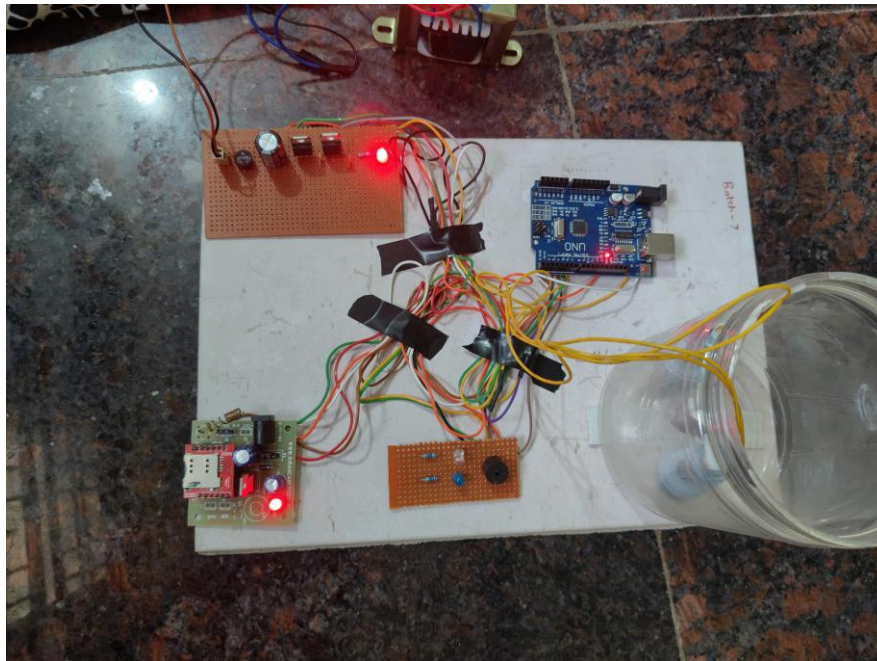


Fig 7.1 circuit is on

The result of the model is a robust and efficient system that enables real-time monitoring of water levels, rainfall, and flow rates. It provides early flood warnings, facilitates timely decision-making, and enhances emergency response capabilities. The system's implementation improves flood management, reduces risks, and enhances public safety in flood-prone areas.

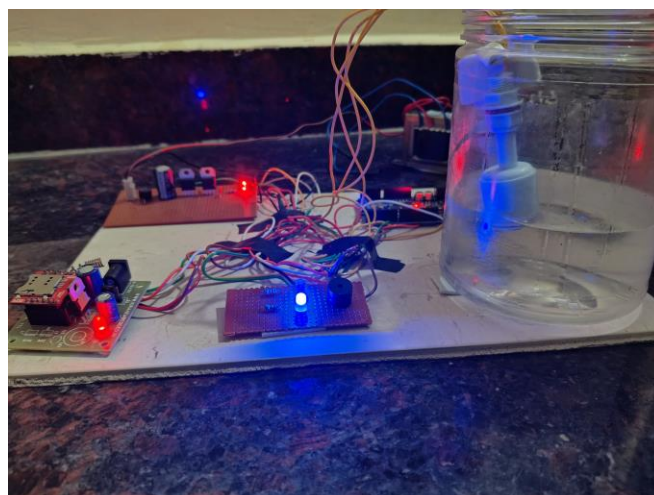


Fig 7.2 first sensor is activated

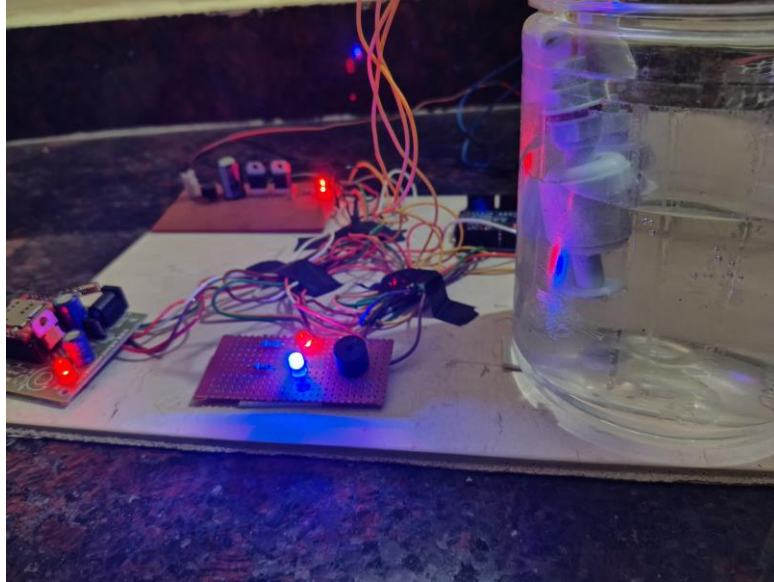


Fig 7.3 second sensor activated



Fig 7.4 alert message sent

CHAPTER-8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Flood alerting and rescue systems play a crucial role in mitigating the impacts of flooding. Their effectiveness hinges on timely detection, efficient communication, and coordinated response efforts. In conclusion, these systems enhance community resilience by providing early warnings that allow for timely evacuations and resource allocation. Integrating advanced technologies, such as satellite imagery and data analytics, with community engagement ensures that alerts reach those most at risk. Continuous training and collaboration among local authorities, emergency services, and community members are essential for improving response times and minimizing casualties. Ultimately, the success of flood alerting and rescue systems depends on a holistic approach that combines technology, preparedness, and community involvement, fostering a safer environment in the face of increasingly frequent and severe flooding events.

8.2 Future scope

The future scope of flood alerting and rescue systems is set to expand significantly through advancements in technology and community engagement. Integration of predictive analytics powered by machine learning will enhance forecasting accuracy, while IoT devices will provide real-time monitoring of water levels and environmental conditions. User-friendly mobile applications will ensure timely alerts reach residents, and community-based solutions will empower local populations to participate in emergency planning and response. Additionally, geospatial technologies like GIS will aid in creating detailed flood risk maps, and drones can be utilized for surveillance and supply delivery. Strengthened inter-agency collaboration and global data sharing will foster a coordinated response to floods. By combining these innovations with education and awareness programs, flood alerting and rescue systems will become more effective, ultimately enhancing resilience and safeguarding communities against future flood risks. Integration of predictive analytics powered by machine learning will enhance forecasting accuracy, while IoT devices will provide real-time monitoring of water levels and environmental conditions. . User-friendly mobile applications will ensure timely alerts reach residents, and community-based solutions will empower local populations to participate in emergency planning and response.

REFERENCE

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2. Billa, L., Mansor, S., & Rodzi Mahmud, A. Spatial information technology in flood early warning systems.
3. Satria, D., Yana, S., Munadi, R., & Syahreza, S. Prototype of Google Maps-Based Flood Monitoring System Using Arduino and GSM Module.
4. Hughes, D., Greenwood, P., Blair, G., Coulson, G., Pappenberger, F., Smith, P., & Beven, K. An intelligent and adaptable grid-based flood monitoring and warning system.
5. Khalid, M. S. B., & Shafiai, S. B. X. Flood disaster management in Malaysia: An evaluation of the effectiveness flood delivery system.

APPENDICES

```
#define L1_Sensor 3
#define L2_Sensor 4

#define BLED 5
#define RLED 6
#define BUZ 13

void setup() {
    Serial.begin(9600);
    pinMode(L1_Sensor, INPUT_PULLUP);
    pinMode(L2_Sensor, INPUT_PULLUP);

    pinMode(BLED, OUTPUT);
    pinMode(RLED, OUTPUT);
    pinMode(BUZ, OUTPUT);

    digitalWrite(BUZ, LOW);
    digitalWrite(BLED, LOW);
    digitalWrite(RLED, LOW);

    Serial.println("AT");
    delay(2000);
    Serial.println("AT+CMGF=1");
    delay(2000);
}

void loop() {
    int L1_st = digitalRead(L1_Sensor);
    int L2_st = digitalRead(L2_Sensor);

    Serial.print(L1_st);
```

```
Serial.println(L2_st);
delay(500);

if ((L1_st == 0) && (L2_st == 0)) {
    digitalWrite(BLED, LOW);
    digitalWrite(RLED, LOW);
    digitalWrite(BUZ, LOW);
}
if ((L1_st == 1) && (L2_st == 0)) {
    digitalWrite(BLED, HIGH);
    digitalWrite(RLED, LOW);
    digitalWrite(BUZ, HIGH);
    delay(500);
    digitalWrite(BUZ, LOW);
}
if ((L1_st == 1) && (L2_st == 1)) {
    digitalWrite(BLED, HIGH);
    digitalWrite(RLED, HIGH);
    digitalWrite(BUZ, HIGH);
    delay(2000);
    digitalWrite(BUZ, LOW);
    Serial.println("AT+CMGS=\"9346352959\"");
    delay(1000);
    Serial.print("Flood Alert ... Hurry up.");
    Serial.write(0x1A);
    delay(2000);
}
}
```